Introduction to Treatment in MSK physiotherapy

Walaa Abu-Taleb

Develop a problem list

1. Pain

- Location (e.g., lower back, shoulder, knee)
- Intensity (mild, moderate, severe)
- Nature (sharp, dull, throbbing)
- Aggravating/relieving factors

2. Range of Motion (ROM)

- Restrictions (e.g., decreased ROM in flexion/extension)
- Asymmetry between sides
- Painful ranges of motion

3. Strength

- Weakness in specific muscle groups
- Functional strength deficits (e.g., inability to lift, carry, squat)

4. Function

- Limitations in daily activities (e.g., walking, climbing stairs)
- Impact on work or sport performance
- Use of assistive devices (e.g., cane, brace)

5. Posture

- Deviations from normal alignment (e.g., forward head, rounded shoulders)
- Effects on biomechanics

6. Swelling/Edema

- Presence and location of swelling
- Impact on movement and function

7. Neurological Symptoms

- Numbness or tingling (paresthesia)
- Weakness in specific areas
- Reflex changes

8. Tissue Integrity

- Skin conditions (e.g., scars, bruising)
- Muscle, tendon, or ligament injuries

9. Patient Education and Compliance

- Understanding of their condition
- Adherence to home exercise programs or activity modifications

10. Psychosocial Factors

- Anxiety or fear related to movement
- Impact of chronic pain on mental health

Example of possible problems

Pain: Right shoulder pain (moderate, sharp, aggravated by overhead activities).
ROM: 120-180° Limited abduction and external rotation in the right shoulder.
Strength: 3/5 strength in right supraspinatus and infraspinatus.
Function: Difficulty lifting objects above shoulder height; limited participation in overhead activities.
Posture: Forward head posture and rounded shoulders observed during standing assessment.
Swelling: Mild swelling noted over the right acromioclavicular joint.
Neurological Symptoms: No neurological deficits reported.
Tissue Integrity: Mild tenderness and swelling in the right rotator cuff area.
Patient Education: Patient lacks understanding of home exercise program; needs education on activity modification.
Psychosocial Factors: Patient expresses anxiety about re-injury and returning to sport.

Bio-psycho-social Model of Functioning, Disability and Health

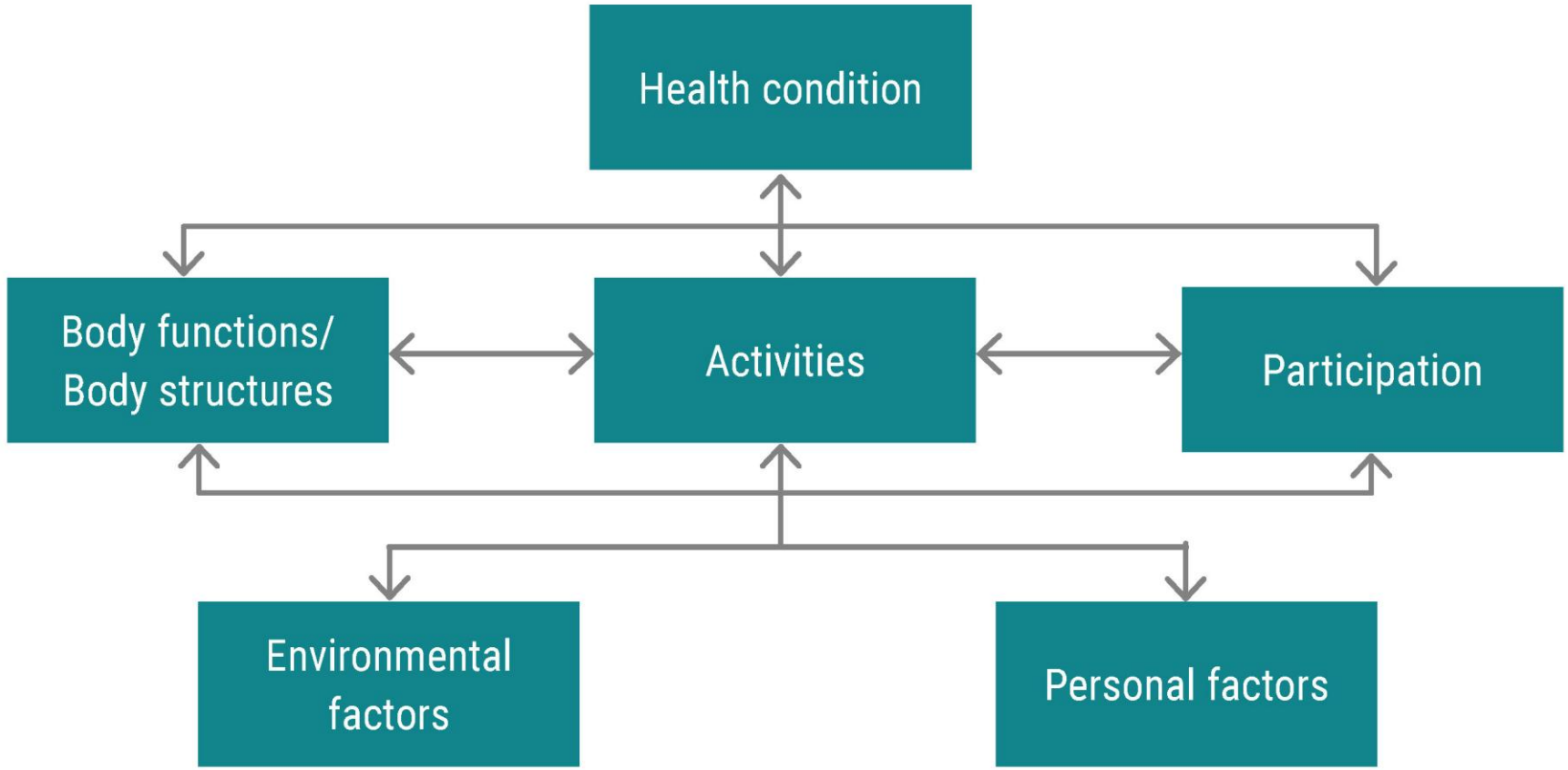


***Figure 1**: Bio-psycho-social model of the International Classification of Functioning, Disability and Health (ICF)*

What does best practice care for musculoskeletal pain look like? Eleven consistent recommendations from high-quality clinical practice guidelines: systematic review

1. Care should be patient centred.
2. Screen patients to identify those with a higher likelihood of serious pathology/red flag conditions.
3. Assess psychosocial factors.
4. Radiological imaging is discouraged unless:
    1. Serious pathology is suspected.
    2. There has been an unsatisfactory response to conservative care or unexplained progression of signs and symptoms.
    3. It is likely to change management.
5. Undertake a physical examination, which could include neurological screening tests, assessment of mobility and/or muscle strength.
6. Patient progress should be evaluated including the use of outcome measures.
7. Provide patients with education/information about their condition and management options.
8. Provide management addressing physical activity and/or exercise.
9. Apply manual therapy only as an adjunct to other evidence-based treatments.
10. Unless specifically indicated (e.g. red flag condition), offer evidence-informed non-surgical care prior to surgery.
11. Facilitate continuation or resumption of work.

Treatment techniques in Physiotherapy

- Pain Management
- Education and Self-Efficacy
- Therapeutic Exercise
- Manual Therapy
- Adjuncts

Pain Management

Physiotherapy is commonly a key aspect in managing chronic pain, being part of this multidisciplinary, bio-psycho-social approach.

Evidence has shown that physiotherapists use a variety of skills to guide and support people with chronic pain towards a better quality of life.

Despite progress in chronic pain management, some patients and some clinicians still expect a ‘cure’. It is often assumed that ‘hands-on’ physiotherapy will be offered, (which is commonly not indicated or appropriate).

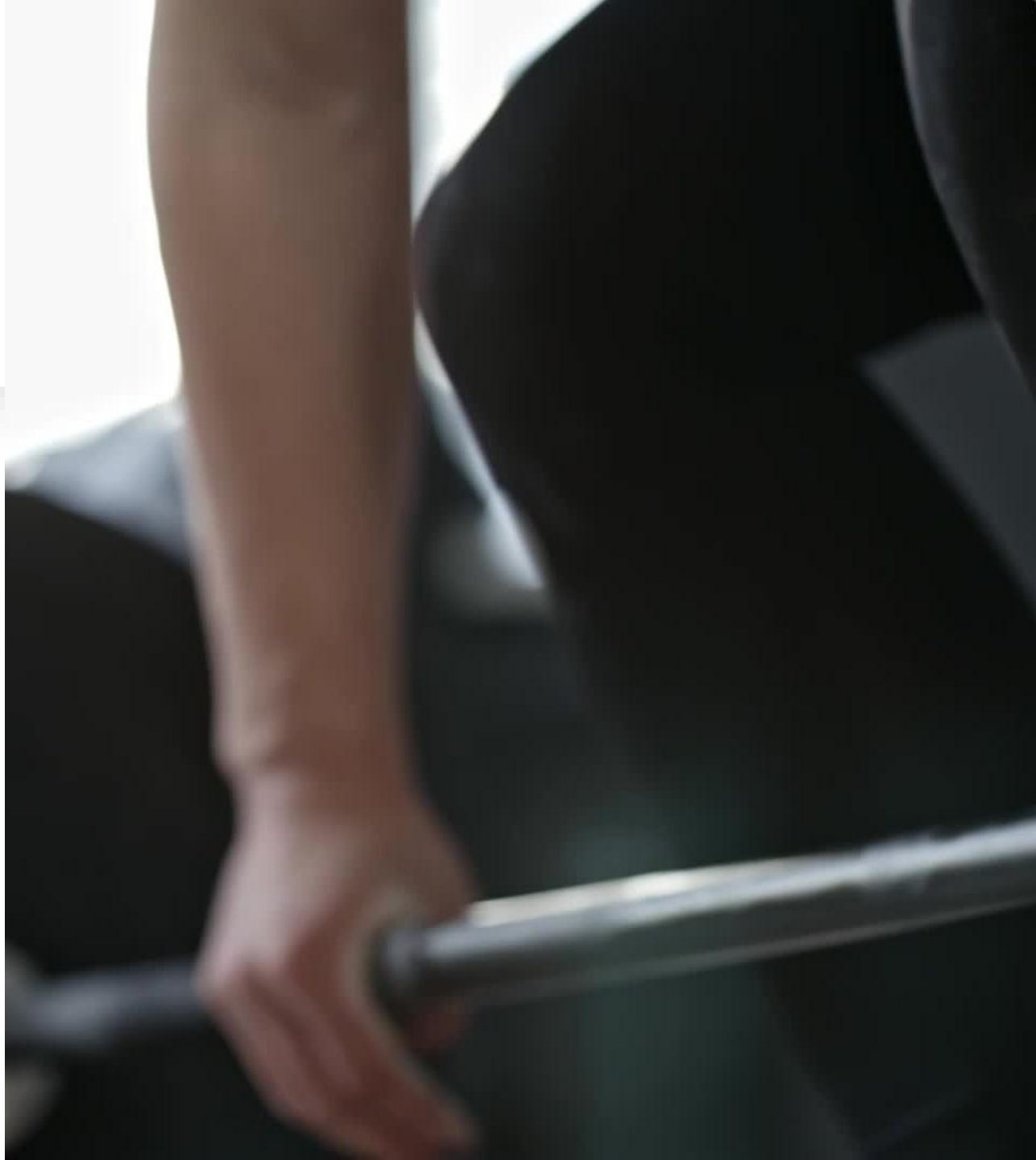
Certain patients fear physiotherapy, exercise and activity, as past experience has worsened their pain or not helped at all.

Pain Management

Intervention	Description
1. Education	General education on injury/disease, acute and chronic pain-physiology, neuroscience education, regimes, activity-modification, ergonomics, general health, coping strategies
2. Sensorimotor training	Can also be termed neuromuscular training or motor control exercises and refers to exercises with the purpose of improving sensorimotor control - including training of muscular synergies, balanced in time and magnitude through the relearning of motor control - in specific, the exercises aim at creating appropriate, automatic and generalized movements to optimize muscular joint stabilization commonly affected by the pain
3. Physical activity-advice	Individualized advice on how to optimize the individual patient's physical activity despite the chronic pain, most often advice on walking, Nordic walking, cross-training or cycling.
4. Interventions aiming at improving structures and functions	Stretching, manual physical therapy techniques, heat or cold treatment, spray-n-stretch-technique, taping – all prior to exercises for movement or stabilization
5. Sensory stimulation	Transcutaneous electrical nerve stimulation, acupuncture
6. Physical activity carried out together with the physiotherapist	Physical activity with direct guidance on for example exercise-bike, cross-trainer or Nordic walking
7. Weight training	Individualized exercises performed in a gym center or weight training with other equipment
8. Relaxation, Mindfulness	Relaxation techniques such as breathing exercises, progressive muscle relaxation, autogenous training. Mindfulness-training including paying attention to thoughts, feelings, bodily sensations other than pain in the present moment, often while carrying out movements
9. Physiotherapy interventions with an Acceptance and commitment therapy (ACT)-inspired approach	Interventions to support acceptance, committed action, values, movement- and behavior change and to reduce fear-of-movement, and also specific interventions on motivation - MI, motivating interviewing - aiming at improving self-efficacy in behavior-change
10. Basic body awareness therapy and training of specified activities	A movement-based physiotherapeutic method developed in Scandinavia where movements aim at enhance body awareness and consciousness of the body with the purpose to move with less effort

Therapeutic Exercise

- Evidence from systematic reviews support the use of therapeutic exercise to improve MSK pain and chronic illnesses in several conditions.
- Incorporation of exercise and PA and their importance are usually introduced during initial individual appointments and continued through to structured exercise classes or as a component of a chronic pain rehabilitation program.
- Using frequency, intensity, type, and time (duration) parameters (F.I.T.T.) for specific exercises.
- The choice of exercise therapy should be closely aligned with patients' preferences and goals as enjoyment in and commitment to the type of exercise will help with long-term adherence



Types of Exercises

1- Strengthening exercise: Endurance (Muscle), strength and power

2- Stretching Exercises

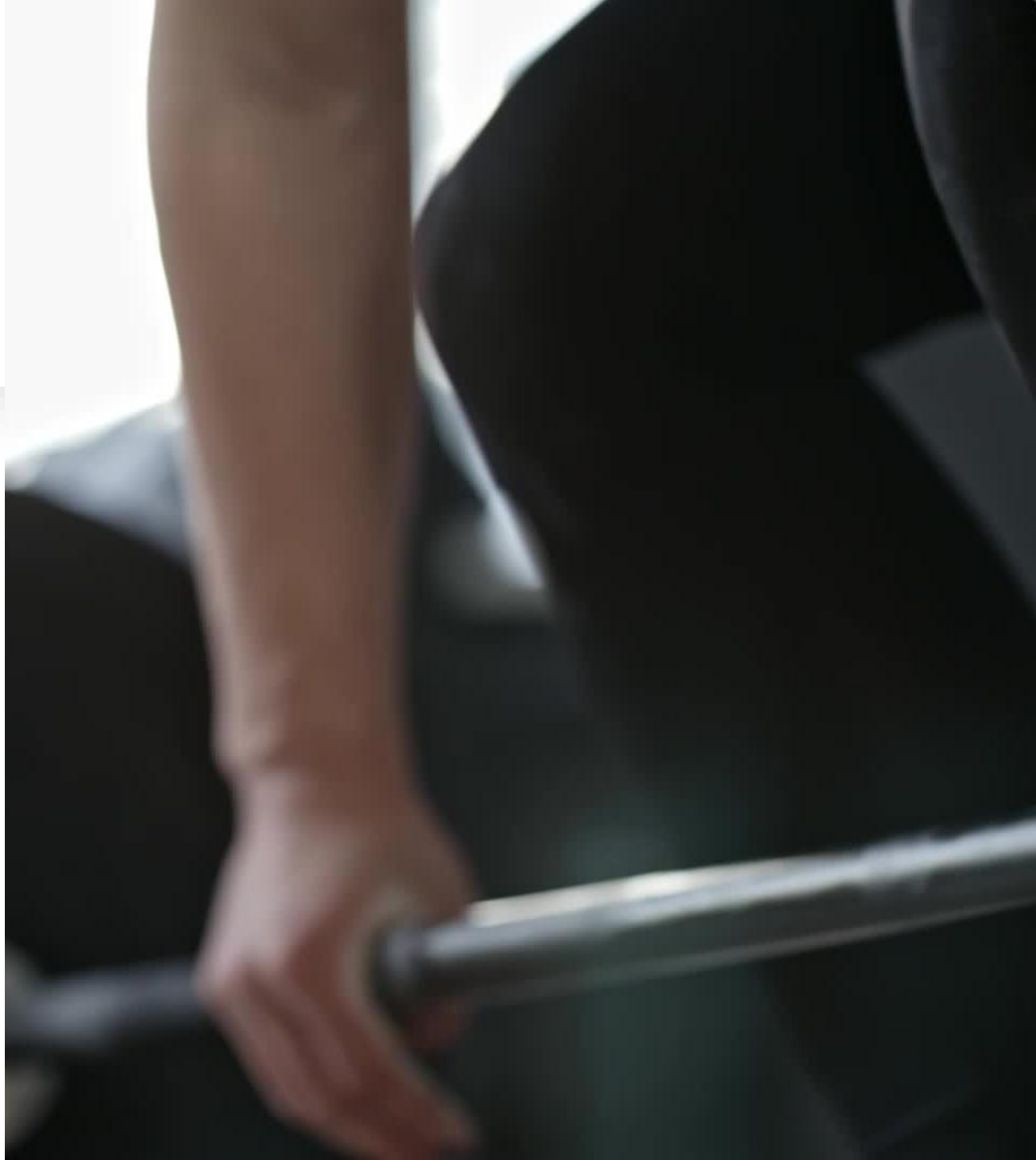
3- Balance Exercises

4- Aerobic Exercises

5- Range of motion Exercises

6- Functional Exercises

7- Perceptual/ Motor Control Exercises



Exercise and function

Functional task	Movement/exercise required
Moving from sitting to standing	Forward flexion of the lumbar spine
Increased standing and walking tolerance	Contracting the quadriceps and gluteal muscle groups
Getting out of bed	Pelvic tilts to mobilize the lumbar spine
Walking to the toilet	Exercising the hamstring and quadriceps muscle groups/balance exercises

Physical Activity and health promotion

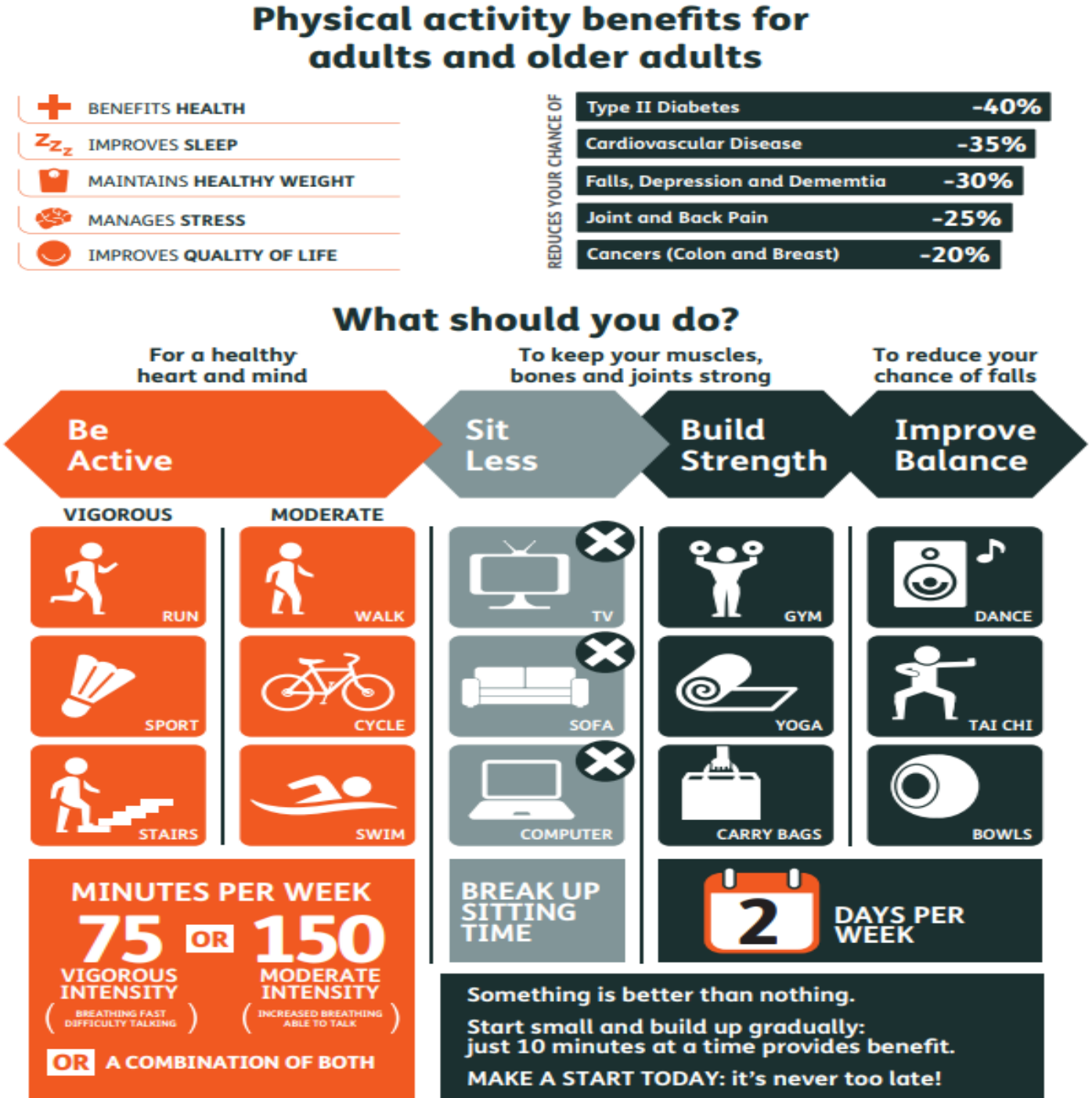
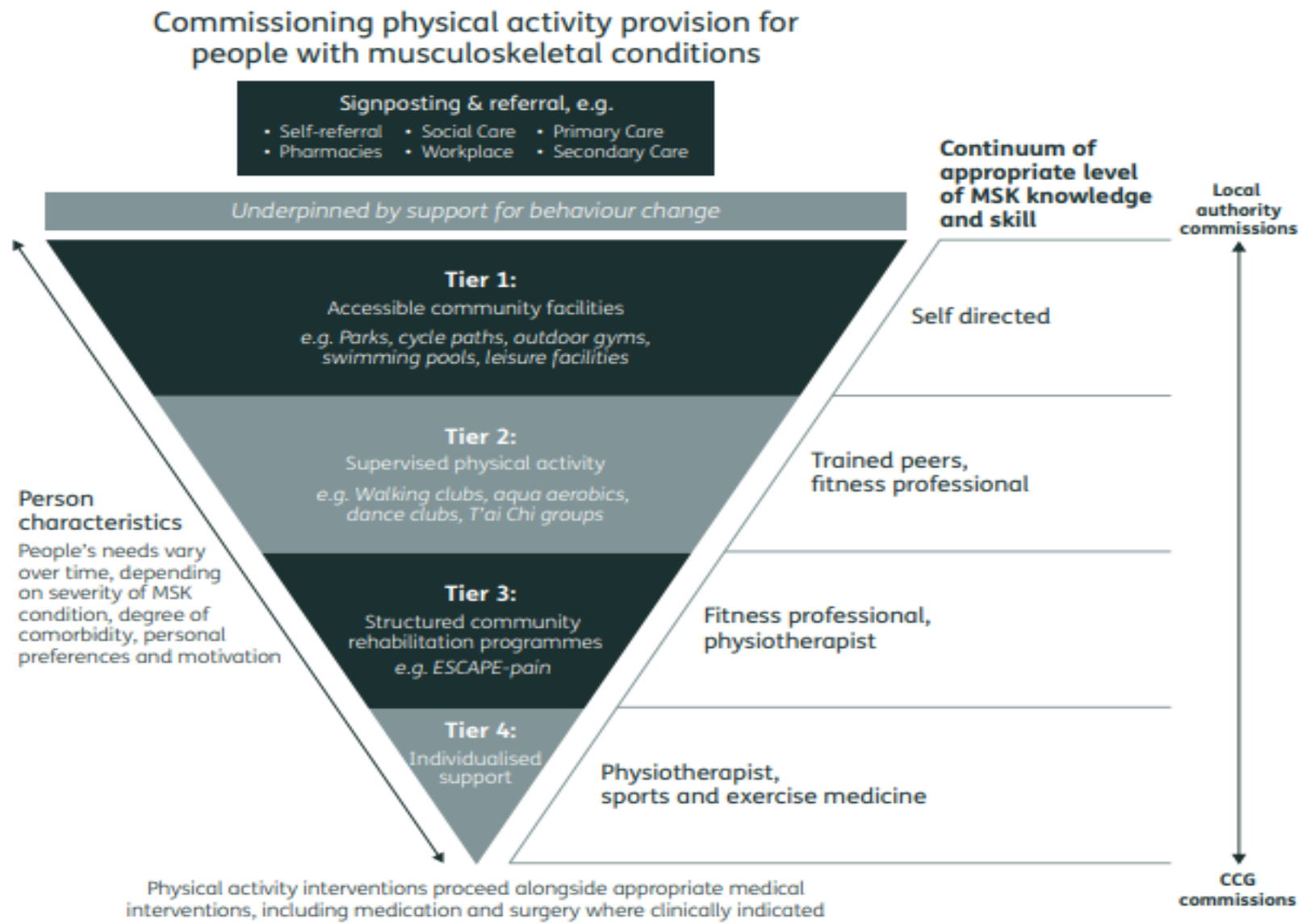


Figure 3: Musculoskeletal physical activity commissioning pyramid



Patient Education

The educational information should aim broadly and include:

- Reassurance and Safety netting
- The aetiology of the pain experience/ their symptoms
- How emotional factors may play a role
- Self-management
- Value of adherence to treatment plan
- How to deal with a daily activities and workload
- Activity modification
- Ergonomic adjustments to perform the limited tasks
- Self-pacing.

Practical Tips for Physical Therapists to Incorporate Self-management Into Practice*

- Establish rapport
- Adopt an active listening communication style
- Provide information
- Develop an action plan
- Goals of treatment are:
  - - Specific
  - - Measurable
  - - Action based
  - - Realistic
  - - Time framed
- Use a nonjudgmental approach
- Facilitate problem solving
- Promote self-efficacy
- Link patients with resources
- Encourage patient to maintain personal health records
- Active follow-up

Adjunctive Therapy

Manual Therapy

Manual therapy is a non-surgical type of conservative management that includes different skilled hands/fingers-on techniques directed to the patient's body (spine and extremities) for the purpose of assessing, diagnosing, and treating a variety of symptoms and conditions.

Manual therapy may be categorised into four major groups:

a) Soft tissue techniques: Trp therapy,	b) mobilisation (non-thrust manipulation),	c) static, and dynamic stretching (Muscle energy technique)	d) Neurodynamic treatment.

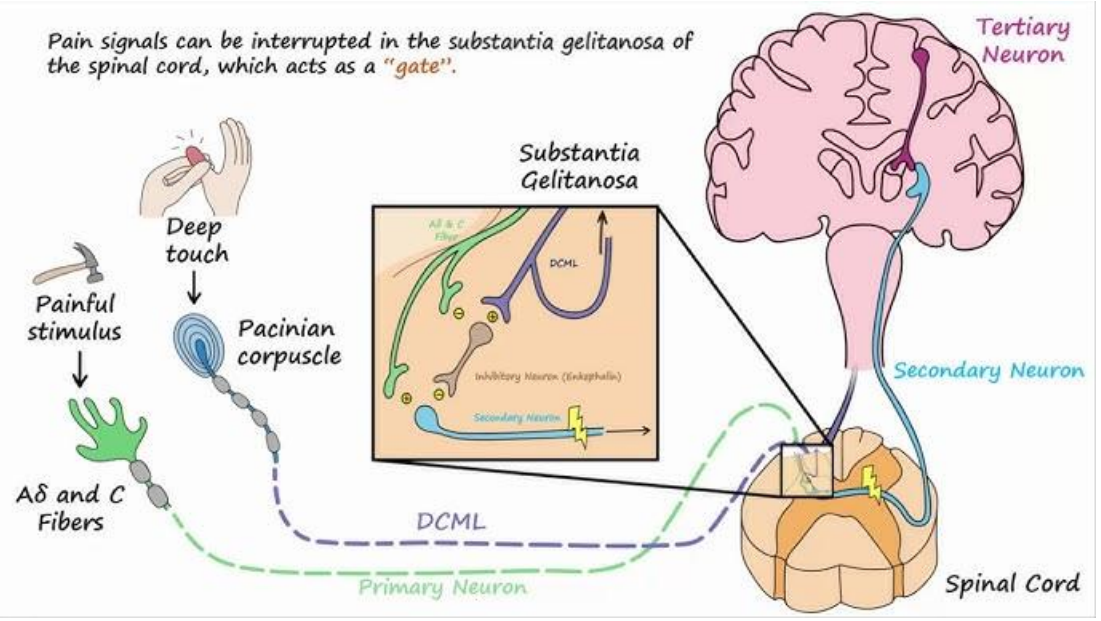
Underlying Mechanism of Efficacy

The rationale has changed from biomechanical paradigm to a neurophysiological one.

Manual therapy may provide its pain-relieving effects via the activation of substantia gelitanosa and descending modulatory pathways in the central nervous system.

Moderately painful pressure can lead to short-lasting pain inhibition (sometimes referred to as pain-inhibits-pain) in both humans and rodents.

Furthermore, manual therapy is likely to have an influence on pain via more cognitive and contextual factors.



Spinal Conditions	Intervention	Evidence		
		Inconclusive	Moderate	High
Acute Low Back Pain	Spinal manipulation /mobilization		positive	
Chronic Low Back Pain	Spinal manipulation /mobilization			positive
Chronic Low Back Pain	Massage		positive	
Chronic Low Back Pain	Foot reflexology added to usual medical care	non-favorable		
Sciatica/ Radiating Leg Pain	Spinal manipulation /mobilization	favorable		
Coccydynia	Spinal manipulation	favorable		
Mid Back Pain	Spinal manipulation	favorable		
Acute/subacute Neck Pain	Thoracic spinal manipulation/ mobilization		positive	
Acute Whiplash-Associated Disorders	Mobilization with exercise		positive	
Chronic Neck Pain	Spinal manipulation /mobilization with exercise		positive	
Neck Pain of any duration	Cervical spinal manipulation/ mobilization alone	favorable		
Chronic Neck Pain	Massage		positive	

Extremity Conditions	Intervention	Evidence		
		Inconclusive	Moderate	High
Shoulder Girdle Pain/ Dysfunction	Manipulation/ mobilization		positive	
Rotator Cuff Pain	Manipulation/ mobilization	favorable		
Shoulder Pain	Massage	favorable		
Adhesive Capsulitis	High-grade mobilization		positive	
Tennis Elbow (Lateral Epicondylitis)	Manipulation	non-favorable		
Tennis Elbow (Lateral Epicondylitis)	Manual tender point therapy	favorable		
Tennis Elbow (Lateral Epicondylitis)	Mobilization with exercise		positive	
Carpal Tunnel Syndrome	Manipulation/ mobilization	favorable		
Hip Osteoarthritis	Manipulation/ mobilization		positive	
Hip Arthroplasty Rehabilitation	Osteopathic manipulative therapy (OMT)	non-favorable		
Knee Osteoarthritis	Manipulation/ mobilization with exercise		positive	
Patellofemoral Pain Syndrome	Manipulation/ mobilization with exercise		positive	
Knee Arthroplasty Rehabilitation	OMT	non-favorable		
Knee Osteoarthritis	Massage	favorable		
Ankle Sprains	Manipulation/ mobilization	favorable		
Ankle Fracture Rehabilitation	Mobilization		negative	
Plantar Fasciitis	Manipulation/ mobilization with exercise		positive	
Morton's Neuroma	Manipulation/ mobilization	favorable		
Hallux Limitus	Manipulation/ mobilization	favorable		
Hallux Abducto Valgus	Manipulation/ mobilization	favorable		

Therapeutic Ultrasound

- Therapeutic US has been previously used by physical therapists on patients with soft-tissue pathologies (with the ability to accurately target deep tissues (3-5 cm)), including tendinitis, bursitis, muscle spasms, neuromas, and myositis ossificans .
- Reports have demonstrated prolonged use of US can cause dose-dependent adverse effects such as burns, soft-tissue necrosis, and hemorrhage.
- There is limited evidence on the clinical efficacy of therapeutic US. US has been shown to have benefits in patients with knee osteoarthritis (OA) when used in addition to laser light therapy and exercise.
- However, studies including soft-tissue conditions such as tendinopathies have shown no benefit from independent US usage, indicating a necessity for further research on its utility in the field of orthopedics.

Transcutaneous Electrical Nerve Stimulation (TENS)

- A technique with few adverse effects, offering the most common application of ENS for the relief of musculoskeletal pain.
- Is applied by surface electrodes offering safe, inexpensive, non-invasive, and easy application.

Mechanism of Action:

- Stimulation of sensory afferent nerve fibers and thereby, enhancement of the pain gate control and opioid system.
- Activation of serotonin, cholinergic and opioid receptors in the spinal cord.

A. Joint Hypomobility (Stiff or Limited Range of Motion)

- **Low-Grade Joint Mobilization**: low-grade (Grade II-III) mobilizations to maintain joint mobility and reduce stiffness.
- **Stretching and ROM Maintenance**: stretching and active range of motion exercises to preserve flexibility.
- **Low-Impact Aerobic Exercises**: Swimming, cycling, or walking to maintain joint health without excessive loading.
- **Strengthening**: surrounding muscle strength with low-resistance exercises to reduce joint load.
- **Hydrotherapy**
- **Adjuncts:** Heat therapy or TENS for localised pain control.

B. Joint Hypermobility

- **Strengthening of Stabilizers** Long-term strengthening programs focused on the muscles surrounding the joint to provide stability.
- **Proprioceptive Training** to improve joint stability and awareness.
- **Functional Training** relevant to the patient's activities to build joint stability in real-world movements.
- **Bracing or Taping** use of braces or taping during activities to prevent further joint damage.
- **Education** joint protection strategies for long-term self-management.

C- Degenerative Joint Disease

- **Activity Modification** as advising patients on modifying high-impact activities to reduce joint stress.
- **Strengthening Exercises** to maintain muscle strength around the joint to reduce the load on the joint.
- **Stretching** of tight soft tissue and active ROM to preserve flexibility.
- **Joint Protection** by educating body mechanics to protect joints during daily activities.
- **Adjuncts**: Use of heat, TENS, or pharmacological interventions (as prescribed) for chronic pain relief. Hydrotherapy low-impact exercises in water to maintain joint movement and reduce pain.

References

- Brona M. Fullen, Harriet Wittink, An De Groef, Morten Hoegh, Joseph G. McVeigh, Denis Martin, Keith Smart, Musculoskeletal Pain: Current and Future Directions of Physical Therapy Practice, Archives of Rehabilitation Research and Clinical Translation, Volume 5, Issue 1, 2023,100258, ISSN 2590-1095, https://doi.org/10.1016/j.arrct.2023.100258. (https://www.sciencedirect.com/science/article/pii/S2590109523000058)
- Bronfort G, Haas M, Evans R, Leininger B, Triano J. Effectiveness of manual therapies: the UK evidence report. Chiropr Osteopat. 2010 Feb 25;18:3. doi: 10.1186/1746-1340-18-3. PMID: 20184717; PMCID: PMC2841070.Clar, C., Tsertsvadze, A., Court, R. et al. Clinical effectiveness of manual therapy for the management of musculoskeletal and non-musculoskeletal conditions: systematic review and update of UK evidence report. Chiropr Man Therap 22, 12 (2014). https://doi.org/10.1186/2045-709X-22-12
- Fullen BM, Wittink H, De Groef A, Hoegh M, McVeigh JG, Martin D, Smart K. Musculoskeletal Pain: Current and Future Directions of Physical Therapy Practice. Arch Rehabil Res Clin Transl. 2023 Feb 1;5(1):100258. doi: 10.1016/j.arrct.2023.100258. PMID: 36968175; PMCID: PMC10036231.
- Trulsson Schouenborg, A., Rivano Fischer, M., Bondesson, E. *et al.* Physiotherapist-led rehabilitation for patients with chronic musculoskeletal pain: interventions and promising long-term outcomes. *BMC Musculoskelet Disord* **22**, 910 (2021). https://doi.org/10.1186/s12891-021-04780-
- Clinical effectiveness of manual therapy for the management of musculoskeletal and non-musculoskeletal conditions: systematic review and update of UK evidence report | Chiropractic & Manual Therapies | Full Text (biomedcentral.com)